

RESEARCH LOGBOOK

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An Instant Depression Screening Method via Valence and Arousal Prediction from Electroencephalogram and Galvanic Skin Response Data via Unsupervised Representation Learning

Materials:

- GSR Device
- EEG Device
- GPU
- Computer
- Data

Day 1. May 5, 2024

- Research on current depression screening methods and limitations
- Collect data on depression status quo

info
• Globally, one in seven 10-19-year-olds experiences a mental disorder, accounting for 13% of the global burden of disease in this age group. • Depression, anxiety and behavioural disorders are among the leading causes of illness and disability among adolescents. • Suicide is the fourth leading cause of death among 15-29 year-olds.
• About 1 in 5 adolescents report symptoms of anxiety or depression, according to a KFF analysis of a new federal survey of teen health. • While some teens are getting mental health care, a significant share say they are not receiving the therapy they need due to costs, fear of what others will think, and/or not knowing how to get help.
• In 2021, 20% of the population ages 12–17 had at least one MDE during the past year. • Among youth ages 12–17, the prevalence of MDE was more than twice as high among females (29%) as among males (12%). • Nearly 60% of youth with MDE in the past year did not receive treatment for depression in 2021.
• The total number of teenagers who recently experienced depression increased 59% between 2007 and 2017. The rate of growth was faster for teen girls (66%) than for boys (44%). • Seven-in-ten U.S. teens said anxiety and depression is a major problem among people their age in the community where they live, according to a Pew Research Center survey of teenagers ages 13 to 17 conducted in fall 2018. An additional 26% cited anxiety and depression as a minor problem.

● According to the National Institute of Mental Health (NIMH), 17% of adolescents aged 12 to 17 had at least one major depressive episode in 2020. That equates to 4.1 million American teens. What's more, 12% of adolescents experienced at least one instance of "severe impairment" that interfered with life. ● The Centers for Disease Control and Prevention (CDC) notes the following worsening trends in mental health in adolescents as of 2021: ○ 42% of students felt persistently sad or hopeless ○ 29% experienced poor mental health ○ 22% considered suicide ○ 10% attempted suicide
● 2021, South Korea recorded a suicide rate of 26 per 100,000 people, the highest among OECD countries. Suicide was the main cause of death for those aged 10 to 39, with 44% of teenage deaths and 56.8% of deaths of those in their 20s attributed to it. Mental illness was identified as the leading cause of suicide.
● Of some 50,000 middle and high school students asked across the nation in 2022, 14.3 percent have thought of suicide over the past 12 months, according to a survey released by the Korea Center for Disease and Prevention Agency on Sunday.
● (Korea) suicide has been the number one cause of death for young people since 2007.
● Approximately 80% to 90% of people with depression who seek treatment eventually respond well to treatment
● The leading cause of disability in the U.S. for ages 15 to 44; MDD can develop at any age, the median age at onset is 32.5 years old. ● MDD affects more than 21 million American adults or about 8.4% of the U.S. population age 18 and older in a given year. ● Almost 49% of adults with MDD are not receiving treatment.
● Number of youth (ages 12-17) with major depression: 4.8 million 19.5% ● Number of youth with severe depression: 3.6 million 14.6% ● 59.8% of youth with depression did not receive any mental health treatment
Rates of adolescent depression increased from 8.1% in 2009 to 15.8% in 2019
To effectively diagnose and treat depression, the doctor must hear about specific symptoms of depression. They may use a series of standard questions to screen for depression. While a physical examination will reveal a patient's overall state of health, by talking with a patient, a doctor can learn about other things that are relevant to making a depression diagnosis. A patient, for example, can report on such things as daily moods, behaviors, and lifestyle habits. A depression diagnosis is often difficult to make because clinical depression can show up in so many different ways. For example, some clinically depressed people seem to withdraw into a state of apathy. Others may become irritable or even agitated. Eating and sleeping patterns can be exaggerated. Clinical depression may cause someone to sleep or eat to excess, or almost eliminate those activities.
To be diagnosed with depression, a person must have symptoms most of the day, nearly every day, for at least 2 weeks. One of the symptoms must be a depressed mood or a loss of interest or pleasure in most activities. Children and adolescents may be irritable rather than sad.

Although several persistent symptoms, in addition to low mood, are required for a depression diagnosis, people with only a few symptoms may benefit from treatment. The severity and frequency of symptoms and how long they last vary depending on the person.

If you think you may have depression, talk to a health care provider, such as a primary care doctor, psychologist, or psychiatrist. During the visit, the provider may ask when your symptoms began, how long they have lasted, how often they occur, and if they keep you from going out or doing your usual activities. It may help to take some notes about your symptoms before the visit.

Certain medications and medical conditions, such as viruses or thyroid disorders, can cause the same symptoms as depression. A provider can rule out these possibilities by doing a physical exam, interview, and lab tests.

Children's Depression Inventory

The Children's Depression Inventory (CDI) is a modification of the Beck Depression Inventory for adults. The CDI is now on its second edition. It assesses depression severity in children and adolescents 7 to 17 years old. Two scales measure emotional problems and functional problems. Three separate rater forms are available: one for parents (17 items), one for teachers (12 items) and a self-report (28 items). Administration time is between five and 15 minutes.

Children's Depression Rating Scale

The Children's Depression Rating Scale (CDRS) was originally designed to measure changes in depressive symptoms in children ages 6 to 12, but its validity and reliability have also been demonstrated in the adolescent population (up to 18 years old). Covering 17 symptom areas, the scale was adapted from the Hamilton Depression Rating Scale and is set up as a semi-structured interview with the child/adolescent. It takes 10 to 15 minutes to administer and score.

Patient Health Questionnaire

The Patient Health Questionnaire (PHQ) is a self-report measure designed to screen depressive symptoms. It takes one to five minutes to complete and roughly the same amount of time for a clinician to review the responses. The PHQ-9 is available in multiple languages.

PATIENT HEALTH QUESTIONNAIRE-9
(PHQ-9)

Over the last 2 weeks, how often have you been bothered by any of the following problems?
(Use "✓" to indicate your answer)

	Not at all	Several days	More than half the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself — or that you are a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed? Or the opposite — being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead or of hurting yourself in some way	0	1	2	3

For office coding: 0 + + +
=Total Score:

If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

Day 2. May 12, 2024

- Develop and send out survey to gather depression-related data

1. Have you tried any forms of depression screening? *

☐ Online, school-based
☐ Online, personally
☐ Depression Center/Clinic
☐ Never (If so, please answer the other questions based on your best knowledge!)

2. On a scale of 1-5, how honest are you during the screenings?

12345

Not honest○ ○ ○ ○ ○Honest

3. If you rated 1-3, what is a possible reason?

☐ I don't want to be seen as "depressed".
☐ Nobody else responds honestly, so I don't as well.
☐ The questions/answer choices are too obvious.
☐ It's too long and annoying.
☐ Other: _____

4. Do you think these depression screening questionnaires are useful/accurate?

☐ Yes
☐ No

5. Do you think there is a negative connotation to depression?

12345

Strongly Disagree○ ○ ○ ○ ○Strongly Agree

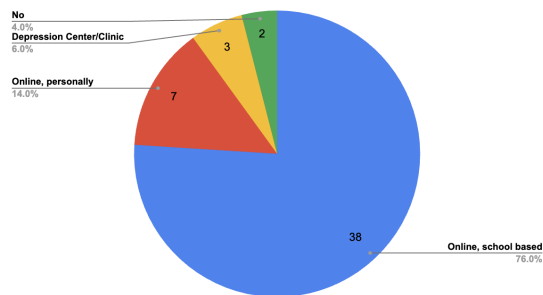
6. Would you trust the results more if technology is incorporated into these depression screening questionnaires?

☐ Yes
☐ No

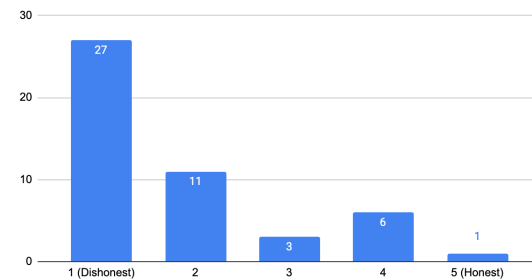
Day 3. May 26, 2024

- Review and analyze survey results
- Review CNN (Convolutional Neural Networks) through research notes (printed out separately)

1. Have you tried any forms of depression screening?

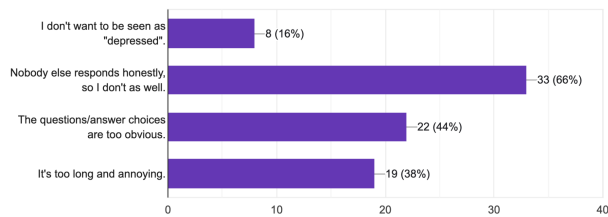


2. On a scale of 1-5, with 1 being dishonest and 5 being honest, how honest are you during the depression screening?



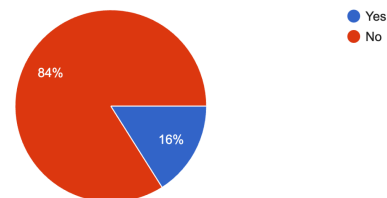
3. If you rated 1-3, what is a possible reason?

50 responses



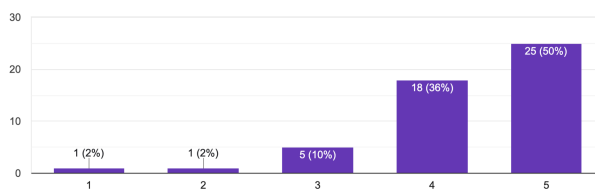
4. Do you think these depression screening questionnaires are useful/accurate?

50 responses



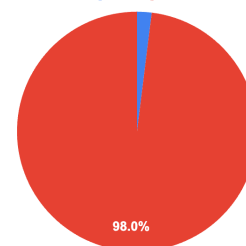
5. Do you think there is a negative connotation to depression?

50 responses



6. Would you trust the results more if technology is incorporated into these depression screening questionnaires?

• Yes • No



Day 4. May 31, 2024

- Study Optimization Techniques (Ch3 GD Algorithm)

Day 5. June 4, 2024

- Begin implementation of CNN model

Day 6. June 9, 2024

- GSR & EEG Background Study
- Galvanic Skin Response (GSR)

- **Definition**
 - Measures the electrical conductivity of the skin in response to stimuli (Science Direct, 2023).
 - Emotional triggers (positive or negative) activate sweat glands, increasing skin conductivity.
- **Physiological Basis**
 - Rapid changes in skin conductance are involuntary and controlled by the autonomic nervous system.
 - Makes GSR effective for detecting emotional intensity without concern for conscious suppression.
- **Measurement Process**
 - The GSR sensor measures electrical conductance between two skin points, typically in microsiemens (μS).
 - Higher conductance signifies higher sweat gland activity and thus higher emotional arousal.
 - Sensors often attach to the index and middle fingers with Velcro; the device is unobtrusive and connects via Bluetooth.
 - Participants are advised to sit still, breathe normally, and avoid conversation to minimize movement artifacts.
- **Signal Components**
 - **Tonic Level:** Slow-varying baseline of skin conductance influenced by ambient conditions and general stress levels.
 - **Phasic Response:** Rapid changes in conductance (peaks/spikes) following emotional stimuli.
- **Data Analysis**
 - The number of phasic responses (“peaks”) is counted over time.
 - A “peak” typically appears 1–5 seconds after exposure to an emotional trigger.
 - More peaks indicate higher levels of emotional arousal (Science Direct, 2023).
- **Electroencephalography (EEG)**
 - **Definition**
 - A non-invasive neurophysiological method detecting electrical activity in the brain via electrodes on the scalp.
 - Useful for identifying abnormalities in brain waves, including seizure activity.
 - **Mechanism**
 - Neurons generate electrical charges through ionic exchange across membranes.
 - These charges propagate through volume conduction, pushing/pulling ions in wave-like patterns.
 - Electrodes pick up voltage differences; signals are amplified and graphed (Tatum Wo et al., 2008).
 - Seizure activity presents as rapid spiking waves on EEG (Johns Hopkins University, 2023).
 - **International 10-20 System**
 - Standardized method for electrode placement based on the underlying cortical regions.

- Each site is labeled with a letter denoting the lobe (Fp, F, T, P, O, C, A) and a number/letter for hemisphere.
- Even numbers = right side; odd numbers = left side; 'z' = midline (Journal of Clinical Neurophysiology, 1991).

Day 7. June 13, 2024

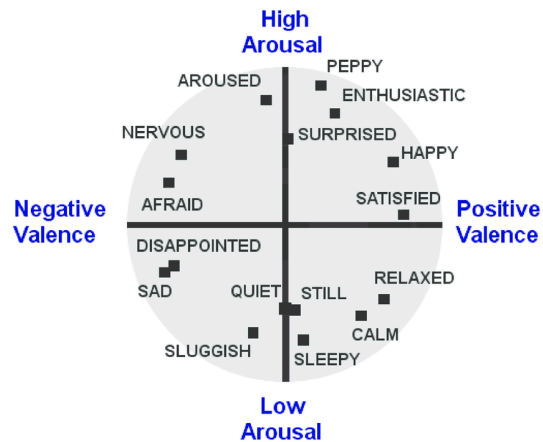
- Gather background knowledge on EEG dataset
- Liu, W., Qiu, J. L., Zheng, W. L., & Lu, B. L. (2021). Comparing recognition performance and robustness of multimodal deep learning models for multimodal emotion recognition. *IEEE Transactions on Cognitive and Developmental Systems*, 14(2), 715-729.
- SEED-IV dataset from Brain-like Computing & Machine Intelligence (BCMI). 72 film clips were chosen and categorized by a preliminary study as having a tendency to induce different emotions, and 15 subjects watched the film clips, while their EEG signals were collected with the 62-channel ESI NeuroScan System. Each session is sliced into 4-second nonoverlapping segments, with one segment regarded as one data sample during the model training.

Day 8. June 17, 2024

- Gather background knowledge on GSR dataset + EEG
- Koelstra, S., Muhl, C., Soleymani, M., Lee, J. S., Yazdani, A., Ebrahimi, T., ... & Patras, I. (2011). Deap: A database for emotion analysis; using physiological signals. *IEEE transactions on affective computing*, 3(1), 18-31.
- DEAP (A Database for Emotion Analysis using Physiological Signals), a multimodal dataset for the analysis of human affective states. The EEG and GSR of 32 participants were recorded as each watched 40 one-minute long excerpts of music videos. Participants rated each video in terms of the levels of arousal, valence, like/dislike, dominance and familiarity.
- Statistical analysis of collected data

Day 9. June 22, 2024

- Research on the Arousal-Valence (A-V) metric



Arousal-Valence Model Examples of Emotions (J. Foucher et al.)

- Describes and categorizes emotions on a two-dimensional graph:
 - **Arousal**: level of physiological and psychological activation (from low to high).
 - **Valence**: positivity or negativity of the emotion (from negative to positive).
- (Neurodivergent Insights, 2024)
- **Arousal Dimension**
 - **Low Arousal**: Emotions such as calm and relaxed.
 - **High Arousal**: Emotions such as excitement or agitation.
- **Valence Dimension**
 - **Negative Valence**: Emotions such as sadness or anger.
 - **Positive Valence**: Emotions such as happiness or delight.
- **Four Quadrants of the Arousal-Valence Model** (Foucher et al.)
 1. **High Arousal, Positive Valence**: Emotions like excitement and surprise.
 2. **High Arousal, Negative Valence**: Emotions such as anger, fear, and anxiety.
 3. **Low Arousal, Positive Valence**: Emotions like calmness and relaxation.
 4. **Low Arousal, Negative Valence**: Emotions like sadness, depression, and boredom.

Day 10. June 26, 2024

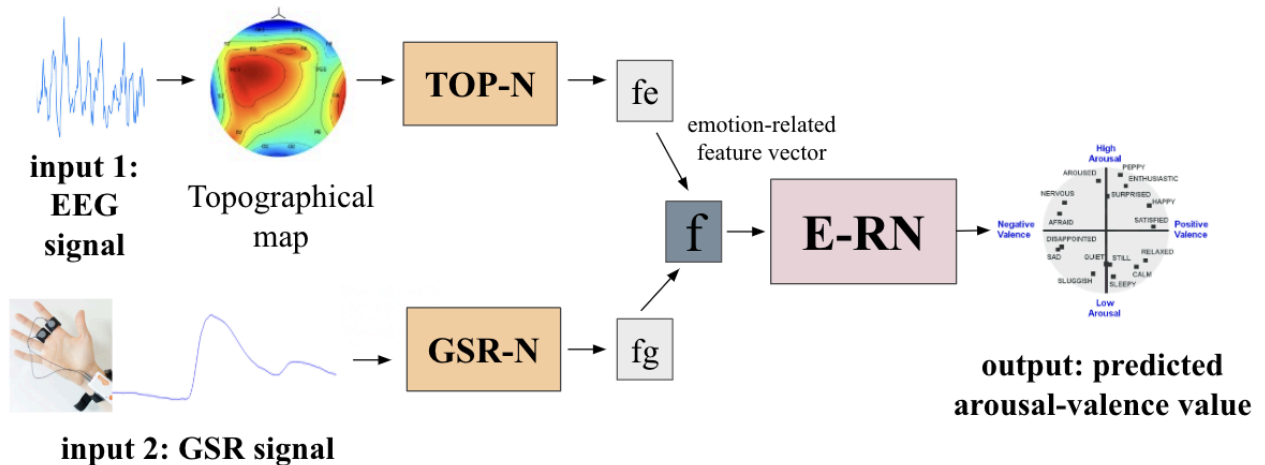
- Review traditional approaches to EEG and GSR-based emotion recognition

Day 11. June 29, 2024

- Conduct a survey on previous research approaches.

Day 12. July 3, 2024

- Develop system overview and draft overview.



Day 13. July 7, 2024

Eq. Cross Entropy Loss Function

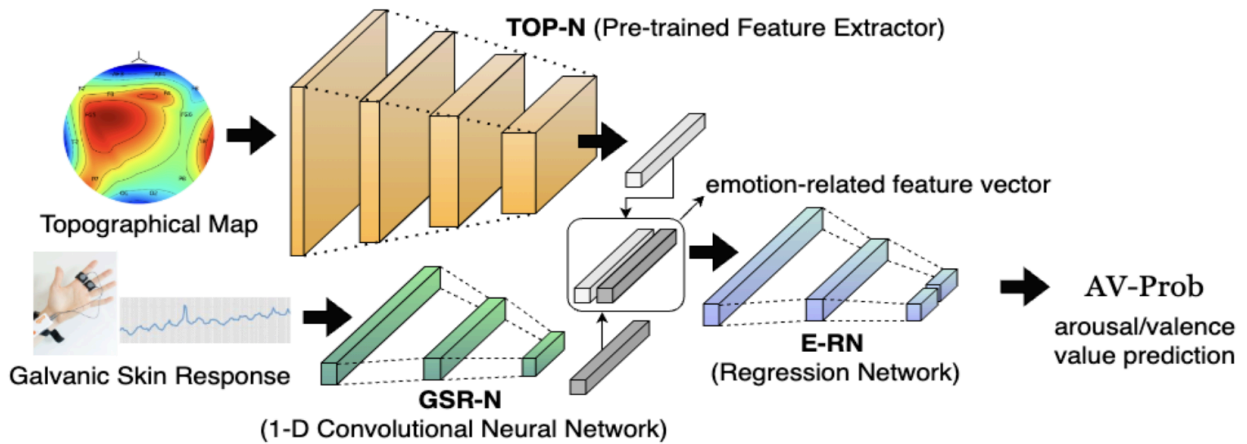
- minimizes difference btwn. similar emotions
- 0: ideal & lowest loss / 1: highest loss

$$L_{av-prob} = \frac{1}{N} \sum_i (arousal_i - \widehat{arousal}_i)^2 + \frac{1}{N} \sum_i (valence_i - \widehat{valence}_i)^2$$

Eq 4. MAE (Mean Absolute Error)

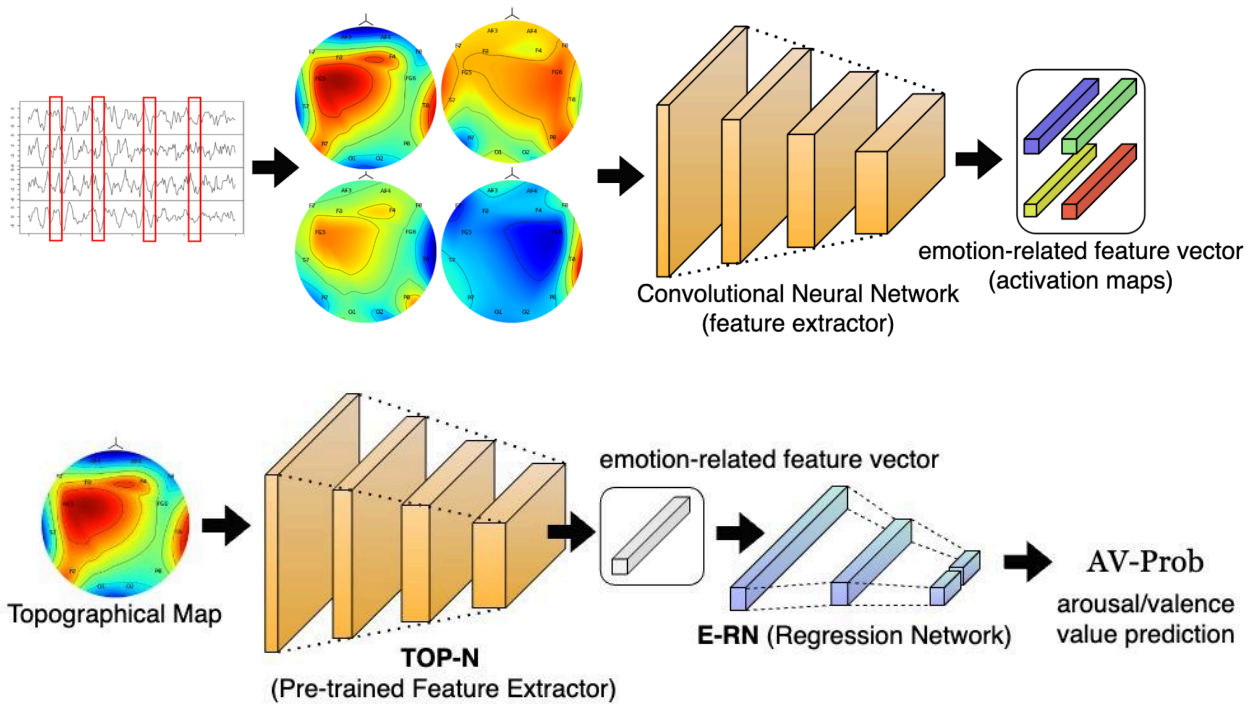
- 0: perfect performance
measures average magnitude of errors between vectors without considering their directions

- Refine proposed system diagram



Day 15. July 16, 2024

- Build TOP-N model component



Day 16. July 21, 2024

- Build GSR-N model component

Day 17. July 25, 2024

- Build E-RN model component

Day 18. July 29, 2024

- Organize and clean dataset for analysis

Day 19. August 3, 2024

- Define and refine evaluation metrics
- 80% random from 2 datasets, 20% for testing

Day 20. August 8, 2024

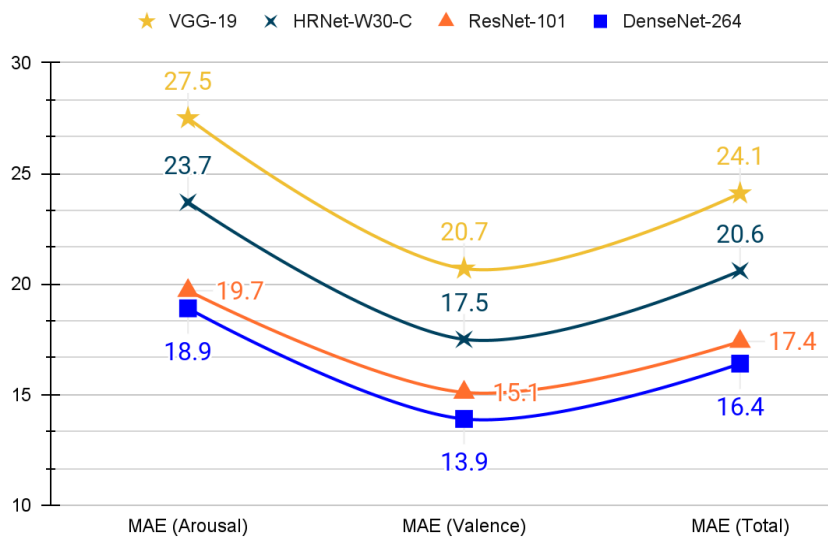
- Train model using representation learning (8-hour session)

Day 21. August 14, 2024

- Train model using transfer learning (1-2 hour session)

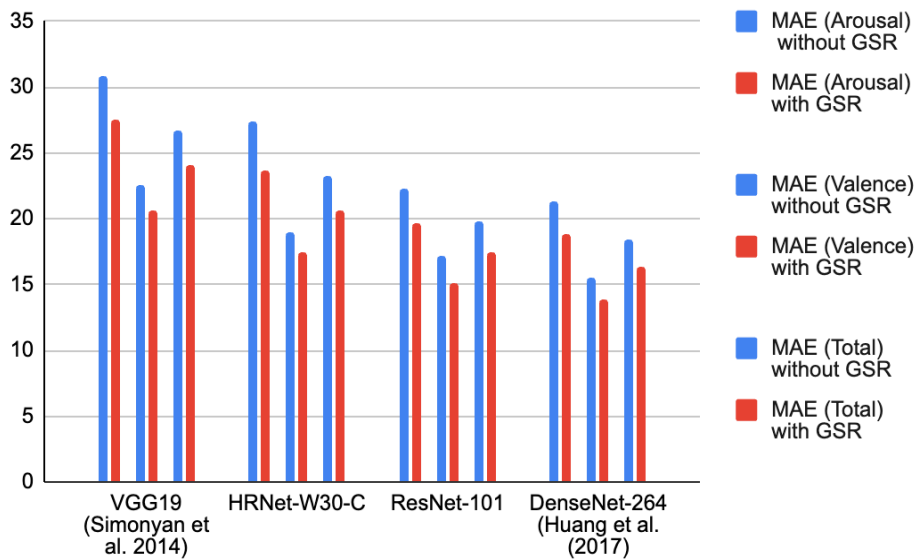
Day 22. August 20, 2024

- Conduct Experiment 1 (MAE network comparison)



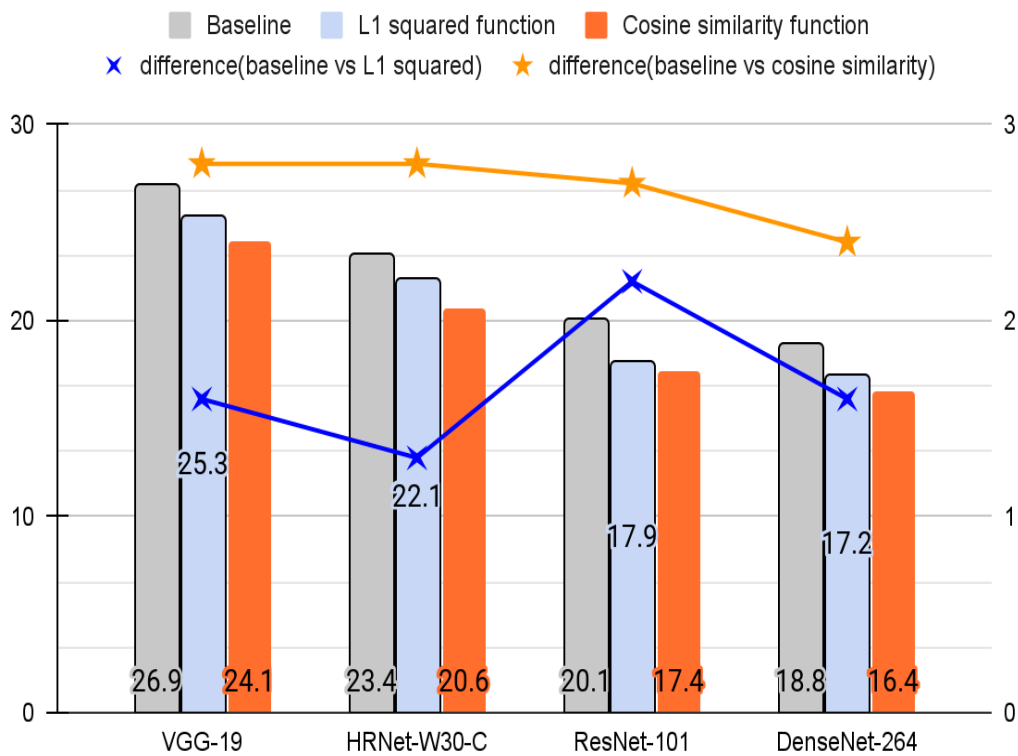
Day 23. August 25, 2024

- Conduct Experiment 2 (MAE GSR comparison)



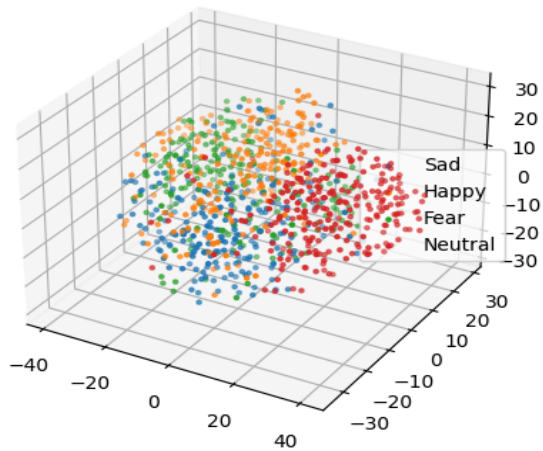
Day 24. August 29, 2024

- Conduct Experiment 3 (Ablation Study)

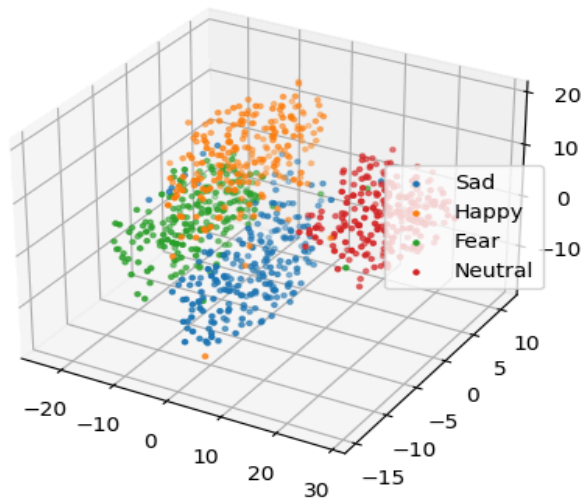


Day 25. September 4, 2024

- Conduct Experiment 4 (t-SNE visualization)

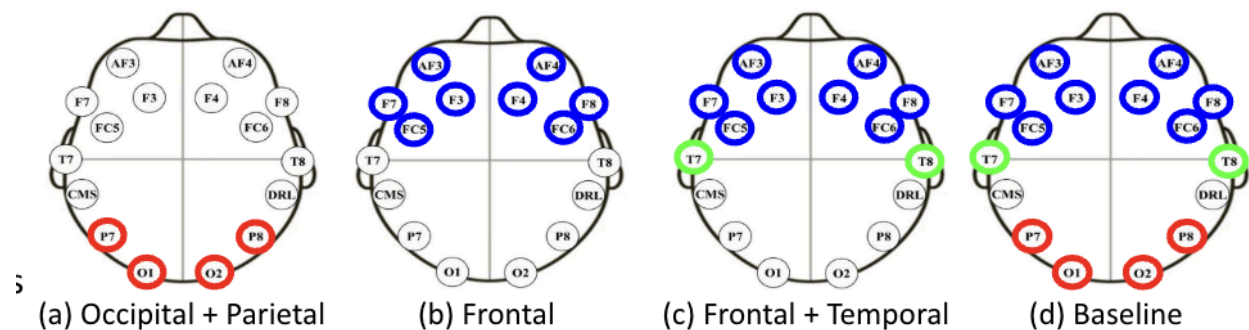


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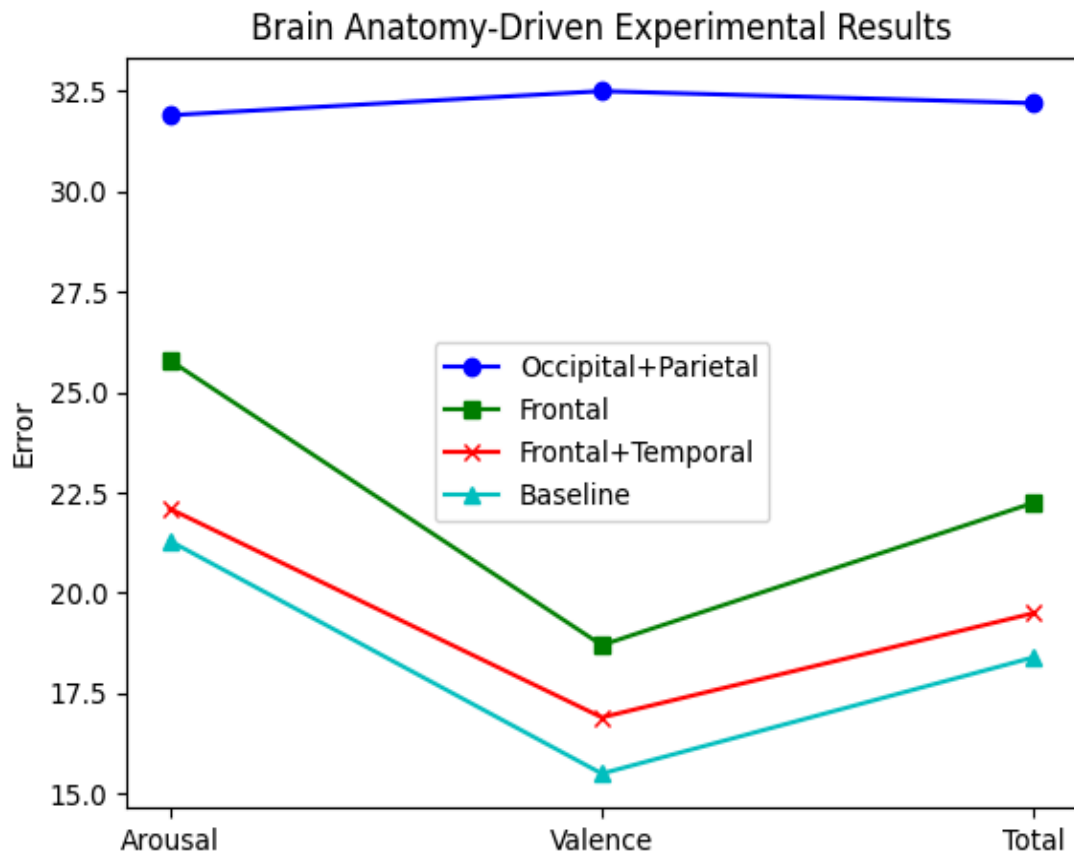
Day 26. September 9, 2024

- Research on brain anatomy related to depression indicators



Day 27. September 14, 2024

- Conduct Experiment 5 (Brain Anatomy Correlation)



- Final data analysis and interpretation